Introduction to Cryptography

Euclidean Algorithm, Extended Euclidean Algorithm and Modular Arithmetic

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Greatest Common Divisor

The greatest common divisor of two integers a and b is written as:

$$\gcd(a, b)$$

It is the largest positive integer that divides both a and b.

Example:

$$60 = 2^2 \times 3 \times 5$$

$$24 = 2^3 \times 3$$

Therefore:

$$\boxed{\gcd(60, 24) = 12}$$

Formal Definition of GCD

A positive integer d is called the greatest common divisor of a and b if:

1. $d \mid a$
2. $d \mid b$
3. Every common divisor of a and b also divides d

Equivalent definition:

$$\boxed{\gcd(a, b) = \max\{k : k \mid a \text{ and } k \mid b\}}$$

GCD and Negative Integers

The signs of the integers do not affect the GCD:

$$\boxed{\gcd(a, b) = \gcd(|a|, |b|)}$$

For example:

$$\gcd(60, 24) = 12$$

$$\gcd(60, -24) = 12$$

$$\gcd(-60, -24) = 12$$

Also:

$$\boxed{\gcd(a, 0) = |a|}$$

Relatively Prime Numbers

Definition

Two integers are called **relatively prime** or **coprime** if their only common positive factor is 1.

Equivalently:

$$\boxed{\gcd(a, b) = 1}$$

Example:

$$\gcd(8, 15) = 1$$

Therefore:

$$\boxed{\text{8 and 15 are relatively prime}}$$

Why are 8 and 15 Relatively Prime?

Positive divisors of 8:

$$1,\ 2,\ 4,\ 8$$

Positive divisors of 15:

$$1,\ 3,\ 5,\ 15$$

The only common divisor is:

$$1$$

Therefore:

$$\boxed{\gcd(8, 15) = 1}$$

Why Do We Need the Euclidean Algorithm?

For small numbers, the GCD can be calculated through prime factorization.
However:

- Prime factorization becomes difficult for large integers.
- Cryptography often involves large numbers.
- We need a faster and simpler method.

Euclidean Algorithm

provides an efficient solution.

Division Algorithm

For integers a and b, where:

$$a > b > 0$$

we can write:

$$\boxed{a = qb + r}$$

where:

$$0 \leq r < b$$

Here:

- a = Dividend
- b = Divisor
- q = Quotient
- r = Remainder

Key Property of the Euclidean Algorithm

Suppose:

$$a = qb + r$$

Then:

$$r = a - qb$$

Any number that divides both a and b must also divide r.
Therefore:

$$\boxed{\gcd(a, b) = \gcd(b, r)}$$

Since:

$$r = a \bmod b$$

we can also write:

$$\boxed{\gcd(a,\ b) = \gcd(b,\ a \bmod b)}$$

Euclidean Algorithm

Starting with:

$$a > b > 0$$

perform repeated division:

$$a = q_1 b + r_1$$

$$b = q_2 r_1 + r_2$$

$$r_1 = q_3 r_2 + r_3$$

$$\vdots$$

until:

$$r_{n-1} = q_{n+1} r_n + 0$$

The final nonzero remainder is:

$$\boxed{\gcd(a, b) = r_n}$$

Euclidean Algorithm: Step-by-Step

1. Divide the larger number by the smaller number.
2. Determine the remainder.
3. Replace:
$$(a, b) \to (b, r)$$
4. Repeat the process.
5. Stop when the remainder becomes 0.
6. The last nonzero remainder is the GCD.

Numerical 1: Find $\gcd(60, 24)$

First division:

$$60 = 2(24) + 12$$

Therefore:

$$\gcd(60, 24) = \gcd(24, 12)$$

Next:

$$24 = 2(12) + 0$$

The last nonzero remainder is 12.
Hence:

$$\boxed{\gcd(60, 24) = 12}$$

Numerical 2: Find $\gcd(48, 18)$

$$48 = 2(18) + 12$$

$$18 = 1(12) + 6$$

$$12 = 2(6) + 0$$

The last nonzero remainder is:

$$6$$

Therefore:

$$\boxed{\gcd(48, 18) = 6}$$

Numerical 3: Find $\gcd(252, 105)$

Step 1:

$$252 = 2(105) + 42$$

Step 2:

$$105 = 2(42) + 21$$

Step 3:

$$42 = 2(21) + 0$$

Therefore:

$$\boxed{\gcd(252, 105) = 21}$$

Numerical 4: Are 35 and 64 Relatively Prime?

$$64 = 1(35) + 29$$

$$35 = 1(29) + 6$$

$$29 = 4(6) + 5$$

$$6 = 1(5) + 1$$

$$5 = 5(1) + 0$$

Therefore:

$$\boxed{\gcd(35, 64) = 1}$$

Hence:

Euclidean Algorithm in Tabular Form

Find:

$$\gcd(252, 105)$$

Dividend	**Divisor**	**Quotient**	**Remainder**
252	105	2	42
105	42	2	21
42	21	2	0

The last nonzero remainder is:

$$\boxed{21}$$

Large Numerical Example

Find:

$$\gcd(1160718174,\ 316258250)$$

Instead of finding the prime factors of these large numbers, we repeatedly use:

$$\boxed{r_{i-2} = q_i r_{i-1} + r_i}$$

until the remainder becomes zero.

Large Numerical Example – Part 1

Dividend	Divisor	Quotient	Remainder
1160718174	316258250	3	211943424
316258250	211943424	1	104314826
211943424	104314826	2	3313772
104314826	3313772	31	1587894
3313772	1587894	2	137984

Large Numerical Example – Part 2

Dividend	Divisor	Quotient	Remainder
1587894	137984	11	70070
137984	70070	1	67914
70070	67914	1	2156
67914	2156	31	1078
2156	1078	2	0

Therefore:

$$\boxed{\gcd(1160718174, 316258250) = 1078}$$

Why is the Euclidean Algorithm Important?

- Efficiently computes the GCD.
- Does not require prime factorization.
- Works with very large integers.
- Determines whether two numbers are relatively prime.
- Forms the basis of the Extended Euclidean Algorithm.
- Is important in public-key cryptography.

Lecture 30: Practice Numericals

Use the Euclidean Algorithm to find:

1. $\gcd(81, 57)$
2. $\gcd(270, 192)$
3. $\gcd(391, 299)$
4. $\gcd(101, 23)$
5. Determine whether 37 and 64 are relatively prime.

Lecture 31

Extended Euclidean Algorithm

Numerical Approach

From Euclidean to Extended Euclidean

The Euclidean Algorithm gives:

$$\boxed{\gcd(a, b)}$$

However, cryptography often requires integers x and y satisfying:

$$\boxed{ax + by = \gcd(a, b)}$$

The procedure used to determine x and y is called the:

Extended Euclidean Algorithm

Bézout's Identity

For integers a and b, there exist integers x and y such that:

$$\boxed{ax + by = \gcd(a, b)}$$

The integers:

$$x,\ y$$

are called:

Bézout Coefficients

Why Extended Euclid Matters in Cryptography

The Extended Euclidean Algorithm is useful for:

- Finding Bézout coefficients.
- Finding modular multiplicative inverses.
- Solving modular equations.
- RSA private-key calculations.
- Checking and working with coprime integers.

Numerical 1: Extended Euclid for 35 and 15

First apply the Euclidean Algorithm:

$$35 = 2(15) + 5$$

$$15 = 3(5) + 0$$

Therefore:

$$\boxed{\gcd(35, 15) = 5}$$

Now we want to express:

$$5 = 35x + 15y$$

Back Substitution

From:

$$35 = 2(15) + 5$$

we obtain:

$$5 = 35 - 2(15)$$

Compare with:

$$5 = 35x + 15y$$

Hence:

$$x = 1$$

and:

$$y = -2$$

Numerical 2: Extended Euclid for 26 and 15

Apply the Euclidean Algorithm:

$$26 = 1(15) + 11$$

$$15 = 1(11) + 4$$

$$11 = 2(4) + 3$$

$$4 = 1(3) + 1$$

$$3 = 3(1) + 0$$

Therefore:

$$\boxed{\gcd(26, 15) = 1}$$

Back Substitution – Step 1

From:

$$4 = 1(3) + 1$$

we obtain:

$$1 = 4 - 3$$

From:

$$11 = 2(4) + 3$$

we obtain:

$$3 = 11 - 2(4)$$

Substitute:

$$1 = 4 - [11 - 2(4)]$$

Back Substitution – Step 2

From:

$$15 = 11 + 4$$

we obtain:

$$4 = 15 - 11$$

Substitute into:

$$1 = 3(4) - 11$$

Therefore:

$$1 = 3(15 - 11) - 11$$

$$1 = 3(15) - 4(11)$$

Back Substitution – Final Step

From:

$$26 = 15 + 11$$

we obtain:

$$11 = 26 - 15$$

Substitute:

$$1 = 3(15) - 4(26 - 15)$$

$$1 = 3(15) - 4(26) + 4(15)$$

$$1 = 7(15) - 4(26)$$

Therefore:

Modular Multiplicative Inverse

The multiplicative inverse of a modulo n is a value a^{-1} such that:

$$\boxed{aa^{-1} \equiv 1 \pmod{n}}$$

The inverse exists only if:

$$\boxed{\gcd(a, n) = 1}$$

The Extended Euclidean Algorithm can be used to calculate the inverse.

Example: Find $15^{-1} \pmod{26}$

From the previous calculation:

$$1 = 7(15) - 4(26)$$

Taking modulo 26:

$$7(15) \equiv 1 \pmod{26}$$

Therefore:

$$\boxed{15^{-1} \equiv 7 \pmod{26}}$$

Check:

$$15 \times 7 = 105$$

$$105 \bmod 26 = 1$$

Numerical 3: Find $7^{-1} \pmod{26}$

Apply Euclidean Algorithm:

$$26 = 3(7) + 5$$

$$7 = 1(5) + 2$$

$$5 = 2(2) + 1$$

$$2 = 2(1) + 0$$

Therefore:

$$\boxed{\gcd(7, 26) = 1}$$

Hence, the inverse exists.

Back Substitution for $7^{-1} \pmod{26}$

From:

$$5 = 2(2) + 1$$

$$1 = 5 - 2(2)$$

Since:

$$2 = 7 - 5$$

we obtain:

$$1 = 5 - 2(7 - 5)$$

$$1 = 3(5) - 2(7)$$

Since:

Result: $7^{-1} \pmod{26}$

We obtained:

$$1 = 3(26) - 11(7)$$

Therefore:

$$-11(7) \equiv 1 \pmod{26}$$

Hence:

$$7^{-1} \equiv -11 \pmod{26}$$

Convert to a positive residue:

$$-11 + 26 = 15$$

Therefore:

$$\boxed{7^{-1} \equiv 15 \pmod{26}}$$

Numerical 4: Find $17^{-1} \pmod{43}$

Euclidean Algorithm:

$$43 = 2(17) + 9$$

$$17 = 1(9) + 8$$

$$9 = 1(8) + 1$$

Therefore:

$$\gcd(17, 43) = 1$$

Hence the inverse exists.

Back Substitution

$$1 = 9 - 8$$

Since:

$$8 = 17 - 9$$

$$1 = 9 - (17 - 9)$$

$$1 = 2(9) - 17$$

Since:

$$9 = 43 - 2(17)$$

$$1 = 2(43) - 5(17)$$

Therefore:

When Does an Inverse Not Exist?

Find:

$$6^{-1} \pmod{15}$$

First calculate:

$$\gcd(6, 15) = 3$$

Since:

$$3 \neq 1$$

the numbers are not relatively prime.
Therefore:

$$\boxed{6^{-1} \pmod{15} \text{ does not exist}}$$

Extended Euclidean Algorithm: Table Method

Initialize:

$$r_0 = a, \qquad r_1 = b$$

$$s_0 = 1, \qquad s_1 = 0$$

$$t_0 = 0, \qquad t_1 = 1$$

Then repeatedly calculate:

$$q_i = \left\lfloor \frac{r_{i-1}}{r_i} \right\rfloor$$

$$r_{i+1} = r_{i-1} - q_i r_i$$

$$s_{i+1} = s_{i-1} - q_i s_i$$

Table Example: Extended Euclid for 26 and 15

q	r	s	t
–	26	1	0
–	15	0	1
1	11	1	-1
1	4	-1	2
2	3	3	-5
1	1	-4	7
3	0	15	-26

At:

$$r = 1$$

we obtain:

$$\boxed{1 = (-4)(26) + 7(15)}$$

Connection with RSA

In RSA, we choose e such that:

$$\gcd(e, \phi(n)) = 1$$

The private exponent d satisfies:

$$\boxed{ed \equiv 1 \pmod{\phi(n)}}$$

Therefore:

$$\boxed{d = e^{-1} \pmod{\phi(n)}}$$

The Extended Euclidean Algorithm is used to calculate d.

Toy RSA-Style Example

Suppose:

$$e = 7$$

and:

$$\phi(n) = 40$$

We need:

$$d = 7^{-1} \pmod{40}$$

Therefore:

$$7d \equiv 1 \pmod{40}$$

Finding $7^{-1} \pmod{40}$

Euclidean Algorithm:

$$40 = 5(7) + 5$$

$$7 = 1(5) + 2$$

$$5 = 2(2) + 1$$

Back substitute:

$$1 = 5 - 2(2)$$

$$1 = 5 - 2(7 - 5)$$

$$1 = 3(5) - 2(7)$$

RSA-Style Example: Result

We obtained:

$$1 = 3(40) - 17(7)$$

Therefore:

$$-17(7) \equiv 1 \pmod{40}$$

So:

$$7^{-1} \equiv -17 \pmod{40}$$

Convert to a positive residue:

$$-17 + 40 = 23$$

Hence:

$$\boxed{d = 23}$$

Classroom Numerical: $17^{-1} \pmod{3120}$

Find:

$$17^{-1} \pmod{3120}$$

Step 1:

$$3120 = 183(17) + 9$$

Step 2:

$$17 = 1(9) + 8$$

Step 3:

$$9 = 1(8) + 1$$

Step 4:

$$8 = 8(1) + 0$$

Back Substitution

$$1 = 9 - 8$$

Since:

$$8 = 17 - 9$$

$$1 = 9 - (17 - 9)$$

$$1 = 2(9) - 17$$

Since:

$$9 = 3120 - 183(17)$$

$$1 = 2(3120) - 367(17)$$

Final Result

We obtained:

$$1 = 2(3120) - 367(17)$$

Therefore:

$$-367(17) \equiv 1 \pmod{3120}$$

Hence:

$$17^{-1} \equiv -367 \pmod{3120}$$

Convert to a positive residue:

$$3120 - 367 = 2753$$

Therefore:

$$\boxed{17^{-1} \equiv 2753 \pmod{3120}}$$

Euclidean vs Extended Euclidean Algorithm

Feature	**Euclidean**	**Extended Euclidean**
Primary output	$\gcd(a, b)$	GCD and coefficients
Equation	GCD only	$ax + by = \gcd(a, b)$
Inverse calculation	No	Yes
RSA use	Check coprimality	Find private exponent

Lecture 31: Practice Numericals

Solve using the Extended Euclidean Algorithm:

1. Express $\gcd(99, 78)$ in the form:
$$99x + 78y$$
2. Find:
$$11^{-1} \pmod{26}$$
3. Find:
$$19^{-1} \pmod{26}$$
4. Determine whether:
$$12^{-1} \pmod{18}$$
exists.
5. Find:
$$23^{-1} \pmod{40}$$

Lecture 32

Modular Arithmetic

Concepts and Numericals

The Modulus

Let a be an integer and n a positive integer.
We define:

$$\boxed{a \bmod n}$$

as the remainder when a is divided by n.
Using the division algorithm:

$$a = qn + r$$

where:

$$0 \leq r < n$$

Therefore:

$$\boxed{a \bmod n = r}$$

The integer n is called the:

Numerical 1: 11 mod 7

Divide 11 by 7:

$$11 = 1(7) + 4$$

Therefore:

$$\boxed{11 \bmod 7 = 4}$$

The remainder is 4.

Numerical 2: $-11 \bmod 7$

We require:

$$0 \leq r < 7$$

Write:

$$-11 = (-2)(7) + 3$$

Therefore:

$$\boxed{-11 \bmod 7 = 3}$$

Important

We normally use the smallest nonnegative residue.

Congruence Modulo n

Two integers a and b are congruent modulo n if:

$$a \bmod n = b \bmod n$$

We write:

$$\boxed{a \equiv b \pmod{n}}$$

For example:

$$73 \equiv 4 \pmod{23}$$

because both produce the same remainder modulo 23.

Equivalent Definition of Congruence

A very useful equivalent definition is:

$$\boxed{a \equiv b \pmod{n}}$$

if and only if:

$$\boxed{n \mid (a - b)}$$

That means:

$$a - b$$

must be exactly divisible by n.

Congruence Examples

Example 1:

$$23 \equiv 8 \pmod 5$$

because:

$$23 - 8 = 15 = 5(3)$$

Example 2:

$$-11 \equiv 5 \pmod 8$$

because:

$$-11 - 5 = -16 = 8(-2)$$

More Examples

$$73 \equiv 4 \pmod{23}$$

because:

$$73 - 4 = 69 = 23(3)$$

Also:

$$21 \equiv -9 \pmod{10}$$

because:

$$21 - (-9) = 30 = 10(3)$$

And:

$$81 \equiv 0 \pmod{27}$$

because:

Special Case: Congruent to Zero

If:

$$a \equiv 0 \pmod{n}$$

then:

$$\boxed{n \mid a}$$

For example:

$$81 \equiv 0 \pmod{27}$$

because:

$$81 = 27(3)$$

Therefore:

$$27 \mid 81$$

Properties of Congruence

Congruence is an equivalence relation.

Reflexive

$$a \equiv a \pmod{n}$$

Symmetric

If:

$$a \equiv b \pmod{n}$$

then:

$$b \equiv a \pmod{n}$$

Transitive

If:

$$a \equiv b \pmod{n}$$

and:

What is Modular Arithmetic?

The operation:

$$a \bmod n$$

maps every integer into:

$$\boxed{\{0, 1, 2, \ldots, n-1\}}$$

For example:

$$Z_8 = \{0, 1, 2, 3, 4, 5, 6, 7\}$$

Arithmetic performed over these residues is called:

Modular Arithmetic

Modular Addition

Property:

$$\boxed{(a + b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n}$$

Example:

$$11 \bmod 8 = 3$$

$$15 \bmod 8 = 7$$

Therefore:

$$(11 + 15) \bmod 8$$

$$= 26 \bmod 8$$

$$\boxed{= 2}$$

Addition Using Reduced Values

Instead of:

$$(11 + 15) \bmod 8$$

reduce each number first:

$$11 \bmod 8 = 3$$

$$15 \bmod 8 = 7$$

Then:

$$(3 + 7) \bmod 8$$

$$= 10 \bmod 8$$

$$\boxed{= 2}$$

Modular Subtraction

Property:

$$\boxed{(a - b) \bmod n = [(a \bmod n) - (b \bmod n)] \bmod n}$$

Example:

$$(11 - 15) \bmod 8$$

$$= -4 \bmod 8$$

$$\boxed{= 4}$$

Similarly:

$$(3 - 7) \bmod 8 = 4$$

Modular Multiplication

Property:

$$\boxed{(ab) \bmod n = [(a \bmod n)(b \bmod n)] \bmod n}$$

Example:

$$(11 \times 15) \bmod 8$$

$$= 165 \bmod 8$$

$$\boxed{= 5}$$

Multiplication Using Reduced Values

Reduce first:

$$11 \bmod 8 = 3$$

$$15 \bmod 8 = 7$$

Then:

$$(3 \times 7) \bmod 8$$

$$= 21 \bmod 8$$

$$\boxed{= 5}$$

This technique becomes very useful with large numbers.

Three Fundamental Modular Rules

Addition:

$$\boxed{[(a \bmod n) + (b \bmod n)] \bmod n = (a + b) \bmod n}$$

Subtraction:

$$\boxed{[(a \bmod n) - (b \bmod n)] \bmod n = (a - b) \bmod n}$$

Multiplication:

$$\boxed{[(a \bmod n)(b \bmod n)] \bmod n = (ab) \bmod n}$$

Modular Exponentiation

Exponentiation can be performed by repeated multiplication and reduction.
Instead of calculating a large value first, we reduce intermediate results.
Example:

$$11^7 \bmod 13$$

We calculate:

$$11^2 \bmod 13$$

then:

$$11^4 \bmod 13$$

and combine these smaller results.

Find $11^7 \bmod 13$: Step 1

First:

$$11^2 = 121$$

Reduce modulo 13:

$$121 \bmod 13 = 4$$

Therefore:

$$\boxed{11^2 \equiv 4 \pmod{13}}$$

Find $11^7 \bmod 13$: Step 2

Now:

$$11^4 = (11^2)^2$$

Using:

$$11^2 \equiv 4 \pmod{13}$$

we obtain:

$$11^4 \equiv 4^2 \pmod{13}$$

$$= 16 \bmod 13$$

$$\boxed{11^4 \equiv 3 \pmod{13}}$$

Find 11^7 mod 13: Final Step

Write:

$$11^7 = 11^4 \times 11^2 \times 11$$

Therefore:

$$11^7 \equiv 3 \times 4 \times 11 \pmod{13}$$

$$= 132 \bmod 13$$

$$\boxed{11^7 \bmod 13 = 2}$$

Residue Set Z_n

The set:

$$\boxed{Z_n = \{0, 1, 2, \ldots, n-1\}}$$

is called the set of residues modulo n.
Examples:

$$Z_4 = \{0, 1, 2, 3\}$$

$$Z_8 = \{0, 1, 2, 3, 4, 5, 6, 7\}$$

Residue Classes Modulo 4

Each element represents an entire residue class.

$$[0] = \{\ldots, -16, -12, -8, -4, 0, 4, 8, 12, 16, \ldots\}$$

$$[1] = \{\ldots, -15, -11, -7, -3, 1, 5, 9, 13, 17, \ldots\}$$

$$[2] = \{\ldots, -14, -10, -6, -2, 2, 6, 10, 14, 18, \ldots\}$$

$$[3] = \{\ldots, -13, -9, -5, -1, 3, 7, 11, 15, 19, \ldots\}$$

Reducing an Integer Modulo n

Finding the smallest nonnegative integer congruent to k modulo n is called:

Reducing k modulo n

Example:

$$23 \bmod 8 = 7$$

Therefore:

$$23 \equiv 7 \pmod 8$$

Also:

$$-1 \equiv 7 \pmod 8$$

Additive Inverse

The additive inverse of x modulo n is a value y such that:

$$\boxed{x + y \equiv 0 \pmod{n}}$$

Example modulo 8:

$$2 + 6 = 8$$

Therefore:

$$\boxed{-2 \equiv 6 \pmod{8}}$$

Similarly:

$$3 + 5 = 8$$

Therefore:

$$\boxed{-3 \equiv 5 \pmod{8}}$$

Additive Inverses Modulo 8

Value	Additive Inverse
0	0
1	7
2	6
3	5
4	4
5	3
6	2
7	1

For example:

$$3 + 5 \equiv 0 \pmod 8$$

Multiplicative Inverse

The multiplicative inverse of x modulo n is a value y satisfying:

$$\boxed{xy \equiv 1 \pmod{n}}$$

We write:

$$y = x^{-1} \pmod{n}$$

Example:

$$3 \times 3 = 9$$

$$9 \bmod 8 = 1$$

Therefore:

$$\boxed{3^{-1} \equiv 3 \pmod{8}}$$

When Does a Multiplicative Inverse Exist?

An integer a has a multiplicative inverse modulo n if and only if:

$$\boxed{\gcd(a, n) = 1}$$

Therefore, a and n must be relatively prime.
For modulo 8:

$$1,\ 3,\ 5,\ 7$$

have inverses because each is relatively prime to 8.

Multiplicative Inverses Modulo 8

x	$\gcd(x, 8)$	$x^{-1} \pmod 8$
1	1	1
2	2	Does not exist
3	1	3
4	4	Does not exist
5	1	5
6	2	Does not exist
7	1	7

Verify the Inverses Modulo 8

$$1 \times 1 \bmod 8 = 1$$

$$3 \times 3 \bmod 8 = 1$$

$$5 \times 5 \bmod 8 = 1$$

$$7 \times 7 \bmod 8 = 1$$

Therefore:

$$\boxed{1^{-1} = 1,\ 3^{-1} = 3,\ 5^{-1} = 5,\ 7^{-1} = 7}$$

modulo 8.

Cancellation in Modular Addition

Suppose:

$$(a + b) \equiv (a + c) \pmod{n}$$

We may cancel a:

$$\boxed{b \equiv c \pmod{n}}$$

Example:

$$5 + 23 \equiv 5 + 7 \pmod{8}$$

Therefore:

$$\boxed{23 \equiv 7 \pmod{8}}$$

Can We Always Cancel in Multiplication?

Suppose:

$$ab \equiv ac \pmod{n}$$

Can we always conclude:

$$b \equiv c \pmod{n}?$$

Can We Always Cancel in Multiplication?

Suppose:

$$ab \equiv ac \pmod{n}$$

Can we always conclude:

$$b \equiv c \pmod{n}?$$

No

Cancellation is valid only if:

$$\boxed{\gcd(a, n) = 1}$$

Example Where Cancellation Fails

Take:

$$a = 6, \qquad n = 8$$

Now:

$$6 \times 3 = 18$$

$$18 \bmod 8 = 2$$

Also:

$$6 \times 7 = 42$$

$$42 \bmod 8 = 2$$

Therefore:

Why Did Cancellation Fail?

Because:

$$\gcd(6, 8) = 2$$

Therefore 6 has no multiplicative inverse modulo 8.

x	0	1	2	3	4	5	6	7
$6x \bmod 8$	0	6	4	2	0	6	4	2

Different values map to the same residue.
Therefore, the mapping is not one-to-one.

Relatively Prime Multiplier

Take:

$$a = 5, \qquad n = 8$$

Since:

$$\gcd(5, 8) = 1$$

calculate:

x	0	1	2	3	4	5	6	7
$5x \bmod 8$	0	5	2	7	4	1	6	3

Every element of Z_8 appears exactly once.

Properties of Modular Arithmetic

For integers in Z_n:

- Addition is commutative.
- Multiplication is commutative.
- Addition is associative.
- Multiplication is associative.
- Multiplication distributes over addition.
- 0 is the additive identity.
- 1 is the multiplicative identity.
- Every element has an additive inverse.
- Only values relatively prime to n have multiplicative inverses.

Euclidean Algorithm Revisited

The Euclidean Algorithm can be expressed as:

$$\boxed{\gcd(a, b) = \gcd(b, a \bmod b)}$$

Example:

$$\gcd(60, 24)$$

Since:

$$60 \bmod 24 = 12$$

we obtain:

$$\gcd(60, 24) = \gcd(24, 12)$$

Euclid Using Mod Notation

Find:

$$\gcd(252, 105)$$

$$252 \bmod 105 = 42$$

Therefore:

$$\gcd(252, 105) = \gcd(105, 42)$$

Next:

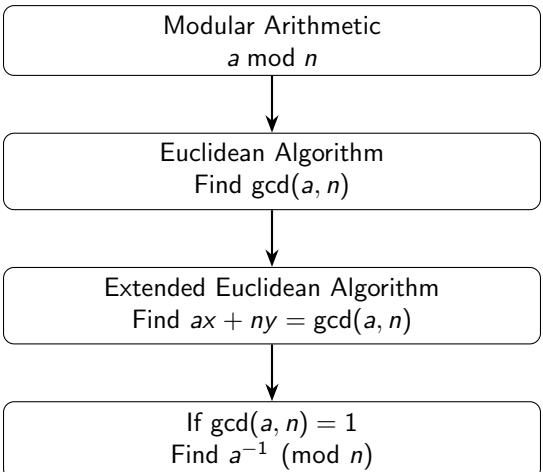
$$105 \bmod 42 = 21$$

Therefore:

$$\gcd(105, 42) = \gcd(42, 21)$$

Finally:

Connecting All Three Concepts



Modular Arithmetic: Practice Numericals

Calculate:

1. $29 \bmod 7$
2. $-15 \bmod 8$
3. $(17 + 25) \bmod 6$
4. $(17 - 25) \bmod 6$
5. $(17 \times 25) \bmod 6$
6. $3^5 \bmod 7$

Practice Answers

$$29 \bmod 7 = \boxed{1}$$

$$-15 \bmod 8 = \boxed{1}$$

$$(17 + 25) \bmod 6 = 42 \bmod 6 = \boxed{0}$$

$$(17 - 25) \bmod 6 = -8 \bmod 6 = \boxed{4}$$

$$(17 \times 25) \bmod 6 = 425 \bmod 6 = \boxed{5}$$

$$3^5 \bmod 7 = 243 \bmod 7 = \boxed{5}$$

Congruence Practice

Determine whether the following are true:

1. $$38 \equiv 3 \pmod{7}$$
2. $$45 \equiv 5 \pmod{10}$$
3. $$-7 \equiv 5 \pmod{12}$$
4. $$56 \equiv 0 \pmod{8}$$

Congruence Practice: Answers

$$38 - 3 = 35 = 7(5)$$

Therefore:

$$38 \equiv 3 \pmod 7$$

$$45 - 5 = 40 = 10(4)$$

Therefore:

$$45 \equiv 5 \pmod{10}$$

$$-7 - 5 = -12$$

Therefore:

$$-7 \equiv 5 \pmod{12}$$

Does $6^{-1} \pmod 8$ Exist?

Calculate:

$$\gcd(6, 8) = 2$$

Since:

$$2 \neq 1$$

the inverse does not exist.
Therefore:

$$\boxed{6^{-1} \pmod 8 \text{ does not exist}}$$

Find $5^{-1} \pmod 8$

We need:

$$5x \equiv 1 \pmod 8$$

Try:

$$x = 5$$

Then:

$$5 \times 5 = 25$$

and:

$$25 \bmod 8 = 1$$

Therefore:

$$\boxed{5^{-1} \equiv 5 \pmod 8}$$

Why Modular Arithmetic Matters in Cryptography

Modular arithmetic is fundamental to:

- RSA
- Diffie–Hellman key exchange
- ElGamal cryptography
- Digital signatures
- Elliptic-curve cryptography
- Modular exponentiation
- Multiplicative inverses
- Finite-field arithmetic

Common Student Mistakes

1. Taking zero as the GCD instead of the last nonzero remainder.
2. Performing back substitution incorrectly.
3. Assuming every integer has a modular inverse.
4. Forgetting:
$$\gcd(a, n) = 1$$
before calculating an inverse.
5. Using a negative remainder instead of the smallest nonnegative residue.
6. Cancelling a multiplier in a congruence without checking coprimality.

Quick Check

Answer briefly:

1. What is $\gcd(a, 0)$?
2. When are two integers relatively prime?
3. State the Euclidean identity.
4. State Bézout's identity.
5. What does $a \bmod n$ represent?
6. What does $a \equiv b \pmod{n}$ mean?
7. When does $a^{-1} \pmod{n}$ exist?
8. Why is Extended Euclid useful in RSA?

Descriptive and Numerical Questions

1. Explain the Euclidean Algorithm with a suitable example.
2. Find $\gcd(270, 192)$ using the Euclidean Algorithm.
3. Explain the Extended Euclidean Algorithm.
4. Find $11^{-1} \pmod{26}$.
5. Explain congruence modulo n with suitable examples.
6. Explain modular addition, subtraction, and multiplication.
7. Explain additive and multiplicative inverses modulo n.
8. Explain why a multiplicative inverse exists only for values relatively prime to the modulus.

Lectures 30–32: Summary

We studied:

- Greatest Common Divisor
- Relatively prime integers
- Euclidean Algorithm
- Extended Euclidean Algorithm
- Bézout's identity
- Modular multiplicative inverse
- Modulus and congruence
- Modular arithmetic operations
- Residue classes Z_n
- Additive and multiplicative inverses
- Modular cancellation
- Cryptographic importance of modular arithmetic

Formula Recap

Division Algorithm:

$$\boxed{a = qn + r, \qquad 0 \leq r < n}$$

Modulus:

$$\boxed{r = a \bmod n}$$

Congruence:

$$\boxed{a \equiv b \pmod{n} \iff n \mid (a - b)}$$

Euclidean identity:

$$\boxed{\gcd(a, b) = \gcd(b, a \bmod b)}$$

Formula Recap: Inverses

Bézout's identity:

$$\boxed{ax + by = \gcd(a, b)}$$

Multiplicative inverse condition:

$$\boxed{a^{-1} \pmod{n} \text{ exists iff } \gcd(a, n) = 1}$$

Definition of modular inverse:

$$\boxed{aa^{-1} \equiv 1 \pmod{n}}$$

RSA-style inverse:

$$\boxed{d = e^{-1} \pmod{\phi(n)}}$$

Questions & Discussion

Modular arithmetic provides the language,

Euclid provides the GCD,

and Extended Euclid provides the inverse.

Your Feedback Matters!

Please scan the QR code and share your feedback about today's lecture.



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Lectures 30–32 – Number Theory and Modular Arithmetic